

Sahaj Raj Malla

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Education

B.Sc. in Computational Mathematics

Dec 2022 – Dec 2026 (expected)

Kathmandu University, Dhulikhel, Nepal

Selected Training & Fellowships

Aspire Leaders Program, Aspire Institute (Harvard University)

Jan 2025 – Mar 2025

AI Fellowship 2023, Fusemachines Nepal

Jan 2023 – Jan 2024

Womanium Quantum 2023, Womanium & QWorld

Jul 2023 – Aug 2023

Data Science Fellowship with DataCamp, Code for Nepal

Dec 2021 – Aug 2023

Annual Nepal AI School (2023, 2024 cohorts), NAAMII

May 2023, Dec 2024

Introduction to Quantum Computing, The Coding School & IBM

Sep 2021 – Apr 2022

Publications

Peer-Reviewed Publications

1. **S. R. Malla** (sole author). [MallaNet: Residual Branch Merge Convolutional Neural Network with Homogeneous Filter Capsules for Devanagari Character Recognition](#). *Scientific Reports (Springer Nature)*, Dec 2025.
99.71% test accuracy on the Devanagari Handwritten Character Dataset, 56.41% fewer parameters than prior SOTA; introduces optimized residual blocks and Homogeneous Filter Capsule layers for multiscale feature extraction.
[Single Author | Q1 Journal]
2. **S. R. Malla** (sole author). [A landmark-based addressing framework for urban navigation using geospatial clustering and pathfinding algorithm](#). *Journal of Science, Engineering and Technology (JSET)*, 19(1), Mar 2025.
37.7% lower average path length and 22.9% lower travel time vs. current systems, using DBSCAN clustering, H3 indexing, and A* search over OpenStreetMap data.
3. B. Kafle, **S. R. Malla**, A. Shrestha, S. Jaiswal (2nd author). [Evaluating XGBoost Algorithm for Nepali Stock Price Prediction](#). *Int. Conf. on Technologies for Computer, Electrical, Electronics & Communication (IEEE)*, Oct 2023.
XGBoost-based prediction model for the Nepali stock market.

Preprints

4. **S. R. Malla** (sole author). [ArthaVyu: A Stacking Ensemble Framework for Multi-Source Real Estate Price Prediction in Nepal](#). *Research Square*, Jun 2026.
Stacking ensemble framework for multi-source real estate price prediction in Nepal; integrates property listings, geospatial data, and market indicators for improved valuation accuracy.
5. **S. R. Malla** (sole author). [Kalimati Vegetable Price Index Forecasting with a Momentum Corrected Online Stacking Ensemble](#). *arXiv:2605.30720*, May 2026.
Built the KVPI, an inverse-volatility-weighted index over 135 daily wholesale commodities (Kathmandu, 2013–2023); tested 14 forecasting models across 7/14/30/90-day horizons. Proposed ensemble reached RMSE 1.771, MAPE 0.68% at 90 days, explaining 84.5% of variance.
6. **S. R. Malla** (sole author). [External Demand, Domestic Monetary Conditions, and Remittance Dynamics in Nepal](#). *arXiv:2605.19401*, May 2026.
ARDL-ECM model of remittances as a share of Nepal's GDP (1993–2024) using PCA-derived demand/monetary indices; error-correction term corrects ~26% of disequilibria annually, projects remittances at ~28.3% of GDP by 2030 under baseline conditions.
7. **S. R. Malla**, S. Kayastha, R. Suwal, H. C. Bhandari, R. Adhikari (1st author). [XGBoost Forecasting of NEPSE Index Log Returns with Walk-Forward Validation](#). *arXiv:2601.08896*, Jan 2026.
Walk-forward validation framework for time-series forecasting of NEPSE index log returns with XGBoost.
8. P. Panth, **S. R. Malla** (2nd author). [Secure, Verifiable, and Scalable Multi-Client Data Sharing via Consensus-Based Privacy-Preserving Data Distribution](#). *arXiv:2601.00418*, Jan 2026.
Consensus-driven privacy-preserving framework for secure, verifiable multi-client data sharing.

9. **S. R. Malla** (sole author). [AI Agents for the Dhumbal Card Game: A Comparative Study](#). *arXiv:2510.11736*, Oct 2025.
Compared AI strategies for the Dhumbal card game; an aggressive heuristic agent reached an 88.3% win rate across 1024 simulations.
10. **S. R. Malla** (sole author). [Devanagari Digit Recognition using Quantum Machine Learning: A Hybrid Quantum-Classical Approach](#). *arXiv:2506.09069*, Jun 2025.
First hybrid quantum-classical architecture for Devanagari digit recognition (CNN + 10-qubit variational circuit); 99.80% test accuracy, 0.9980 average per-class F1.

Professional Experience

Founder <i>HamroNiwas, Banepa, Nepal</i>	Nov 2023 – Present
Built a rental and marketplace platform for Nepali students, connecting 4,000+ users and reducing search time by 90%.	
Data Consultant <i>YoungInnovations Pvt. Ltd., Kathmandu, Nepal</i>	May 2025 – Jun 2025
Designed and optimized ELT pipelines with Apache Spark for large-scale dataset processing; automated workflows with Apache Airflow. (Contract.)	
Data Scientist <i>Naamche Technology, Kathmandu, Nepal</i>	Sep 2021 – Jan 2023
Scraped and analyzed 1M+ real estate listings (Florida, Texas, California); produced investment-decision reports, improving decision accuracy by 40+% and speeding report computation roughly threefold.	
Data Engineer <i>Devfinity Nepal, Kathmandu, Nepal</i>	Mar 2021 – Sep 2021
Built a Django-based data factory processing 1M+ job records daily, deployed on Azure.	
Founder <i>vyesna.com, Kathmandu, Nepal</i>	Apr 2020 – Mar 2021
Built a Django/React job platform with a scraped-data recommender system.	

Teaching & Volunteering

Teaching Assistant <i>NAAMII AI School, Kathmandu, Nepal</i>	Dec 2024 – Jan 2025
Lead Quantum Computing Researcher (Volunteer) <i>OneQuantum, Munich (Remote)</i>	Mar 2021 – Mar 2023
Applied the Quantum Approximate Optimization Algorithm (QAOA) to Max-Cut formulations for network configuration problems.	
Website Manager / Executive Member <i>Kathmandu University Robotics Club (KURC), Dhulikhel, Nepal</i>	May 2023 – May 2024
Data Scientist / Software Developer (Volunteer) <i>Robotics Association of Nepal (RAN), Kathmandu, Nepal</i>	Mar 2021 – Mar 2022

Conference Talks & Panels

Conference Presentation, Nepali Stock Price Prediction <i>IEEE, Bhaktapur, Nepal</i>	Oct 2023
Conference Presentation, Landmark-Based Addressing <i>Kathmandu University, Dhulikhel, Nepal</i>	Oct 2024
Panelist, National Youth Scientific Conference 2024 <i>National Youth Council Nepal, Kathmandu, Nepal</i>	Jun 2024
Guest Speaker, Career in AI/ML <i>CSIT Association of Nepal, Pokhara, Nepal</i>	Dec 2023

Selected Projects

[DistilBERT, Nepali News Classification](#)

Aug 2022

Associated with IT MEET 2022

DistilBERT sequence-classification model over 9 Nepali newsgroup datasets; 95.475% accuracy.

[Solving Human Trafficking on a Quantum Computer](#)

Nov 2021 – Apr 2022

Associated with OneQuantum

Applied the Quantum Approximate Optimization Algorithm and the Max-Cut problem to find optimal node configurations.

[GreatAPI](#)

Jun 2022 – Present

Personal project, open-source Python web framework

Full-stack FastAPI framework; self-built and maintained, used as the basis for a training workshop (Sep 2023).

[SahajMails](#)

Nov 2025 – Present

Personal project, open-source Python library

Personalized bulk email tool for outreach, newsletters, and campaigns; secure, cost-free email automation.

[Mission Oxygen Web Portal](#)

May 2021 – Jun 2022

Associated with Robotics Association of Nepal

Facilitated access to vaccines, ventilators, oxygen cylinders, and hospital details during the COVID-19 pandemic.

Honors & Awards

Winner, Infinity Codewave 2026 Hackathon, Kathmandu University	May 2026
Winner, Infinity Data Visualization Hackathon, Kathmandu University	May 2026
Winner, Computer Engineering Model Demonstration, KEC LITE 2081, Kantipur Engineering College	Dec 2024
Winner, KU Hackfest 2023 (OSM Category), Kathmandu University Computer Club	Oct 2023
Winner, Seeds for the Future Nepal, Huawei Technologies Co., Ltd.	Sep 2023
Winner, Infinity Codewave, Department of Mathematics, Kathmandu University	Aug 2023
Winner, AI Competition (Text Challenge), IT MEET 2022, Kathmandu University	Aug 2022
Winner, AI Competition (Data Visualization), IT MEET 2022, Kathmandu University	Aug 2022
Winner, Youth Innovation Challenge 2022, Social Changemakers and Innovators	Aug 2022

Press & Media

Making Devanagari Handwriting Accessible: The Story of MallaNet , <i>Business 360°</i>	Feb 2026
Kathmandu University Student Builds an AI Model to Recognize Handwritten Devanagari Script , <i>Techpana</i> (in Nepali)	Dec 2025
South Asia's 1st Quantum AI For Devanagari Digits – 99.8% Accuracy Breakthrough , <i>ICT Frame</i>	Jun 2025
Innovation on Home Soil: The Paths of Possibility Shown by Sahaj , <i>Tera Online</i> (in Nepali)	Apr 2025
A Journey with “Seeds for the Future 2023” – A Technological Odyssey from Nepal to China , <i>ICT Frame</i>	Nov 2023

Technical Skills

Deep Learning: PyTorch; CNN, residual, and capsule architectures; parameter-efficient model design; transformer fine-tuning (DistilBERT)

Agents & RL: LangGraph; agent simulation and evaluation; game environments and strategy benchmarking

Quantum ML: Qiskit, PennyLane; variational quantum circuits; hybrid quantum-classical models

Forecasting & Data: XGBoost, walk-forward validation, stacking ensembles, ARDL-ECM; Apache Spark, Apache Airflow

Engineering: Python, C++, R, TypeScript; Docker, AWS, Azure; Django, FastAPI

Mathematics: Optimization, numerical methods, probability & statistics (B.Sc. Computational Mathematics)